화학컴퓨터 (2009년 1학기)

시험 #2 정답

(총 **70점**)

1. **(총 6 점, 각 3 점)**

(a) $1+\sqrt{3}i$ 에서

$$r=\sqrt{1^2+(\sqrt{3})^2}=2\text{이고, }\tan\theta=\frac{\sqrt{3}}{1}=\sqrt{3}\Rightarrow\theta=\frac{\pi}{3}$$

이므로, $1+\sqrt{3}i$ 을 복소평면에서 직교좌표의 극좌표 변환을 이용하여 변환하면 다음과 같다.

$$1+\sqrt{3}i=2\Big(\cos\Big(\frac{\pi}{3}\Big)+i\sin\Big(\frac{\pi}{3}\Big)\Big)$$

Euler 공식을 이용하면,

$$1+\sqrt{3}i=2\Big(\cos\Big(\frac{\pi}{3}\Big)+i\sin\Big(\frac{\pi}{3}\Big)\Big)=2e^{i\pi/3}$$

그러므로,

$$\begin{aligned}(1+\sqrt{3}i)^{12} &= \Big(2e^{i\pi/3}\Big)^{12}=(2)^{12}e^{i(\pi/3)\times 12}\\ &= 2^{12}\Big(\cos\Big(4\pi\Big)+i\sin\Big(4\pi\Big)\Big)\\ &= 2^{12}.\end{aligned}$$

(b)

$$\begin{aligned}(1-i)e^{-i\pi/6} &= (1-i)\Big(\cos\Big(-\frac{\pi}{6}\Big)+i\Big(-\frac{\pi}{6}\Big)\Big)\\ &= (1-i)\Big(\frac{\sqrt{3}}{2}-\frac{1}{2}i\Big)\\ &= \frac{\sqrt{3}}{2}-\frac{1}{2}i-\frac{\sqrt{3}}{2}i-\frac{1}{2}\\ &= \Big(\frac{\sqrt{3}}{2}-\frac{1}{2}\Big)-\Big(\frac{\sqrt{3}}{2}+\frac{1}{2}\Big)i\end{aligned}$$

2. **(총 8 점, 각 2 점)**

(a)

$$\begin{aligned}\frac{dy}{dx} &= -2x^{-3}(1-e^x)\sin(3x)+x^{-2}(-e^x)\sin(3x)+x^{-2}(1-e^x)3\cos(3x)\\ &= -2x^{-3}(1-e^x)\sin(3x)-x^{-2}e^x\sin(3x)+3x^{-2}(1-e^x)\cos(3x).\end{aligned}$$

(b)

$$\begin{aligned}\frac{g'(x)}{g(x)} &= -\frac{A}{x^2} + \ln x + 1 \\ g'(x) &= g(x)\Big(-\frac{A}{x^2} + \ln x + 1\Big) \\ &= \exp\left(\frac{A}{x} + x\ln x\right)\Big(-\frac{A}{x^2} + \ln x + 1\Big)\end{aligned}$$

$(g(x) = A/x + x\ln x \Rightarrow g(x) = e^{A/x + x\ln x})$

(c)

$$\begin{aligned}\frac{du}{dr} &= A\big(-12r^{-13}\big) - B\big(-6r^{-7}\big) \\ &= -\frac{12A}{r^{13}} + \frac{6B}{r^7}.\end{aligned}$$

(d)

$$\frac{dy}{d} = -\frac{e^x}{1-e^x}.$$

3. **(총 8 점, 각 4점)**

(a)

$$\begin{aligned}f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= 2a\cos(bx)(-\sin(bx))b = -2ab\cos(bx)\sin(bx) \\ f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= -3b\sin^2(y)(\cos(y)) = -3b\sin^2(y)\cos(y)\end{aligned}$$

(b)

$$\begin{aligned}f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= \frac{a(c+xy) - a(x+by)y}{(c+xy)^2} = \frac{ac - aby^2}{(c+xy)^2} \\ f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= \frac{ab(c+xy) - a(x+by)x}{(c+xy)^2} = \frac{abc - ax^2}{(c+xy)^2}\end{aligned}$$

4. **(총 8 점, 각 4점)**

(a)

$$
\begin{aligned}
f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= 2axe^{ax^2+by^2} \\
f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= 2bye^{ax^2+by^2} \\
f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)_y = 2ae^{ax^2+by^2} + 4a^2x^2e^{ax^2+by^2} \\
f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)_x = 2be^{ax^2+by^2} + 4b^2y^2e^{ax^2+by^2} \\
f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y \partial x}\right) &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y = 4abxye^{ax^2+by^2} \\
f_{yx}(x,y) = \left(\frac{\partial^2 f}{\partial x \partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x = 4abxye^{ax^2+by^2}
\end{aligned}
$$

(b)

$$
\begin{aligned}
f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y = e^{-y}\big(-y\sin(xy)\big) &= -ye^{-y}\sin(xy) \\
f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x = -e^{-y}\big(-x\sin(xy)\big) &= -e^{-y}\cos(xy) - xe^{-y}\sin(xy) \\
f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)_y &= -y^2e^{-y}\cos(xy) \\
f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)_x &= e^{-y}\cos(xy) + xe^{-y}\sin(xy) + xe^{-y}\sin(xy) - x^2e^{-y}\cos(xy) \\
&= e^{-y}\cos(xy) + 2xe^{-y}\sin(xy) - x^2e^{-y}\cos(xy) \\
f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y \partial x}\right) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y &= -e^{-y}\sin(xy) + ye^{-y}\sin(xy) - xye^{-y}\cos(xy) \\
&= e^{-y}\big(-\sin(xy) + y\sin(xy) - xy\cos(xy)\big) \\
f_{yx}(x,y) = \left(\frac{\partial^2 f}{\partial x \partial y}\right) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x &= ye^{-y}\sin(xy) - e^{-y}\sin(xy) - xye^{-y}\cos(xy) \\
&= e^{-y}\big(-\sin(xy) + y\sin(xy) - xy\cos(xy)\big)
\end{aligned}
$$

5. **(총 4점, 각 2점)**

$$\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT \Rightarrow \boxed{P = \frac{nRT}{V-nb} - \frac{n^2a}{V^2}}$$

0|므로

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{nRT}{(V-nb)^2} + \frac{2n^2a}{V^3}.$$

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{nR}{V-nb}.$$

(a)

$$\frac{\partial}{\partial T}\left(\frac{\partial P}{\partial V}\right)_T = \frac{\partial}{\partial T}\left[-\frac{nRT}{(V-nb)^2}+\frac{2n^2a}{V^3}\right] = -\frac{nR}{(V-nb)^2}.$$

(b)

$$\frac{\partial}{\partial V}\left(\frac{\partial P}{\partial T}\right)_V = \frac{\partial}{\partial V}\left[\frac{nR}{V-nb}\right] = -\frac{nR}{(V-nb)^2}.$$

6. **(총 12 점, 각 3점)**

(a) $df(x,y) = by\cos(bx)dx + \sin(bx)dy \Rightarrow$ 완전미분, $f(x,y) = y\sin(bx) + C.$

(b) $df(x,y) = y\ln x dx + x\ln y dy \Rightarrow$ 불완전미분.

(c) $df(x,y) = x\cos x dx + y\sin y dy \Rightarrow$ 완전미분 $f(x,y) = x\sin x + \cos x - y\cos y + \sin y + C.$

(d) $df(x,y) = (2x/y)dx - (x^2/y^2)dy \Rightarrow$ 완전미분, $f(x,y) = x^2/y + C.$

7. **(총 12점, 각 4점)**

(a) $f(x,y)$가 (x^*,y^*)에서 극값을 가지므로

$$\begin{aligned} f_x(x^*,y^*) &= 0 \Rightarrow 2x^* + 2 = 0 \Rightarrow x^* = -1 \\ f_y(x^*,y^*) &= 0 \Rightarrow 2y^* = 0 \Rightarrow y^* = 0 \end{aligned}$$

(b) Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*,y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*,y^*)f_{yy}(x^*,y^*) - f_{xy}(x^*,y^*)^2.$$

주어진 함수 $f(x,y) = x^2 + y^2 + 2x$에 대하여,

$$\begin{aligned} f_{xx}(x,y) &= 2 \\ f_{yy}(x,y) &= 2 \\ f_{xy}(x,y) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = 2 > 0 \quad \text{and} \quad \Delta_2 = 2\times 2 - 0^2 = 4 > 0$$

$\Delta_1 > 0$이고 $\Delta_2 > 0$이므로 $f(x,y)$는 $x^* = 2,\ y^* = 0$에서 극소값을 갖는다.

(c) Lagrange 미정승수법을 이용하면

$$f_x(x^*, y^*) = \lambda g_x(x^*, y^*) \quad \Rightarrow \quad 2x^* + 2 = -\lambda \Rightarrow x^* = -\frac{\lambda}{2} - 1 \tag{1}$$

$$f_y(x^*, y^*) = \lambda g_y(x^*, y^*) \quad \Rightarrow \quad 2y^* = \lambda \Rightarrow y^* = \frac{\lambda}{2} \tag{2}$$

제한조건 $g(x, y) = y - x = 0$ 과 (1) 식과 (2) 식으로부터

$$y^* - x^* = 0 \Rightarrow \tfrac{\lambda}{2} - \Big(-\tfrac{\lambda}{2} - 1 \Big) = 0$$
$$\lambda = -1$$

그러므로

$$x^* = -\frac{1}{2} \quad y^* = -\frac{1}{2}$$

8. **(총 12점, 각 3점)**

(a) 온도와 부피에 대한 내부에너지 $U(T, V)$ 의 전미분은 다음과 같이 구할 수 있다.

$$dU = \Big(\frac{\partial U}{\partial T}\Big)_V dT + \Big(\frac{\partial U}{\partial V}\Big)_T dV. \tag{3}$$

(b) 온도와 부피에 대한 엔트로피 $S(T, V)$ 의 전미분은 다음과 같이 구할 수 있다.

$$dS = \Big(\frac{\partial S}{\partial T}\Big)_V dT + \Big(\frac{\partial S}{\partial V}\Big)_T dV. \tag{4}$$

(c) (b) 로부터

$$\begin{aligned} dU &= TdS - PdV \\ &= T\bigg(\Big(\frac{\partial S}{\partial T}\Big)_V dT + \Big(\frac{\partial S}{\partial V}\Big)_T dV \bigg) - PdV \qquad ((3) \text{로부터}) \\ &= T\Big(\frac{\partial S}{\partial T}\Big)_V dT + \bigg(T\Big(\frac{\partial S}{\partial V}\Big)_T - P \bigg) dV \end{aligned} \tag{5}$$

(d) (a) 와 (c) 의 결과를 비교하면 (3) 식으로부터

$$dU = \Big(\frac{\partial U}{\partial T}\Big)_V dT + \Big(\frac{\partial U}{\partial V}\Big)_T dV. \tag{6}$$

그리고 (5) 식으로부터

$$dU = T\Big(\frac{\partial S}{\partial T}\Big)_V dT + \bigg(T\Big(\frac{\partial S}{\partial V}\Big)_T - P \bigg) dV \tag{7}$$

그러므로

$$\begin{aligned} \Big(\frac{\partial U}{\partial T}\Big)_V &= T\Big(\frac{\partial S}{\partial T}\Big)_V \\ \Big(\frac{\partial U}{\partial V}\Big)_T &= T\Big(\frac{\partial S}{\partial V}\Big)_T - P. \end{aligned}$$